

BACHELOR OF EDUCATION
MAS 101: DESCRIPTIVE STATISTICS

Instructions: Attempt **QUESTION 1** and any other **TWO** questions.

Question 1 [Compulsory – 30 marks]

a) In a survey carried out among users of Maseno Health Centre about the quality of the service provided, the following responses were recorded: 30 said that the service was very good, 20 said it was good, 8 said it was average, 2 said it was bad and 1 very bad. **[5 marks]**

- i. Indicate the population under study.
- ii. Indicate the sample size.
- iii. Indicate the statistical unit.
- iv. Indicate the variable under study.
- v. Classify the measurement level of the variable under study.

b) Mention the main phases (steps) of statistical research and describe just one of them. **[5 marks]**

c) For the values (scores) 205, 6, 5, 5, 5, 2 and 1, which measure of central tendency is appropriate to describe this set of scores? Why? **[3 marks]**

d) The personnel department of a certain firm carried out a survey of the salaries of 20 workers in the administrative sector, having obtained the following results.

Salary range (KSh 1,000.00)	[0-2]	[2-4]	[4-6]	[6-8]	[8-10]
Frequency	0.05	0.20	0.35	0.25	0.15

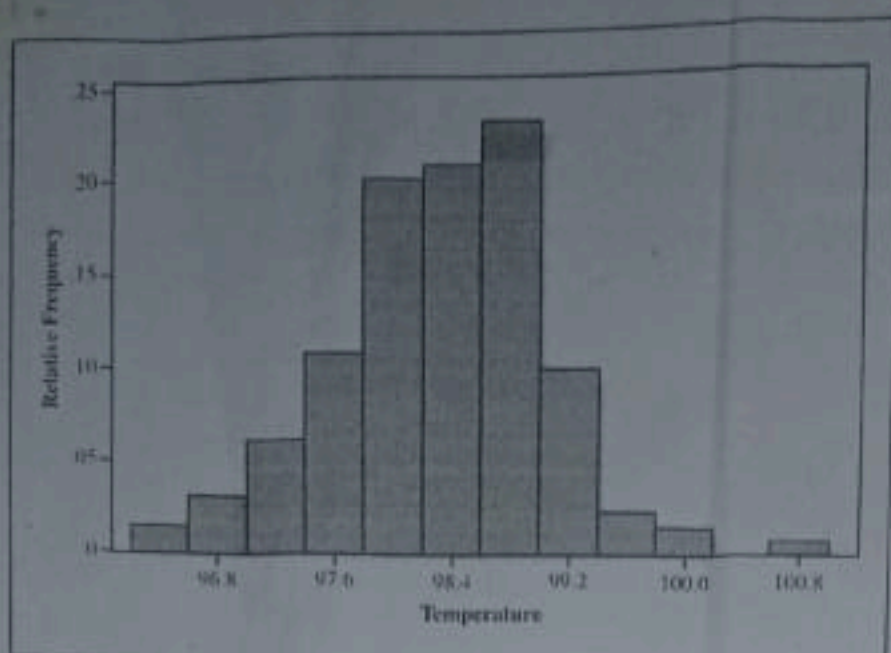
i. Calculate approximately the mean, variance and standard deviation of salaries. **[6 marks]**

ii. If a 100% increase is granted to all employees, will there be a change in average salaries? And what about the variation in salaries? Justify. **[3 marks]**

iii. Classify the salary distribution regarding asymmetry. **[3 marks]**

e) Pearson's coefficient of skewness for a distribution is 0.4 and its coefficient of variation is 30%. If its mode is 88, find the mean and median. **[3 Marks]**

f) Graphs are useful for illustrating data distributions, but they have limitations. Identify any two of these limitations. **[2 Marks]**



- Describe the shape of the distribution of temperatures. [3 Marks]
- Are there any unusual observations? Can you think of any explanation for these? [3 Marks]
- Locate the 98.6-degree standard on the horizontal axis of the graph. Does it appear to be near the centre of the distribution? [3 Marks]

b) A researcher is investigating the relationship between advertising expenditure (in thousands of Kenya shillings) and sales revenue (in millions of Kenya shillings) for a company over 12 months. The simple linear regression model for this relationship is given as:

$$\hat{Y} = 1.55 + 0.12X$$

Where:

$\hat{Y}$ represents the predicted sales revenue (in millions of dollars)

X represents advertising expenditure (in thousands of dollars)

Using this regression model, answer the following questions:

- What do the coefficients of the regression model represent in the context of the study? [3 Marks]
- Predict the sales revenue if the company spends Ksh 60,000 on advertising. [2 Marks]
- Calculate the residual if the actual sales revenue for an advertising expenditure of Ksh 35,000 was Ksh 5.5 million. Explain what the residual represents. [2 Marks]
- If the coefficient of determination (R^2) for the model is 0.95, explain what this means in terms of the relationship between advertising expenditure and sales revenue. [2 Marks]
- Interpret the scatter plot of the model [2 Marks]

Question 2 [20 marks]

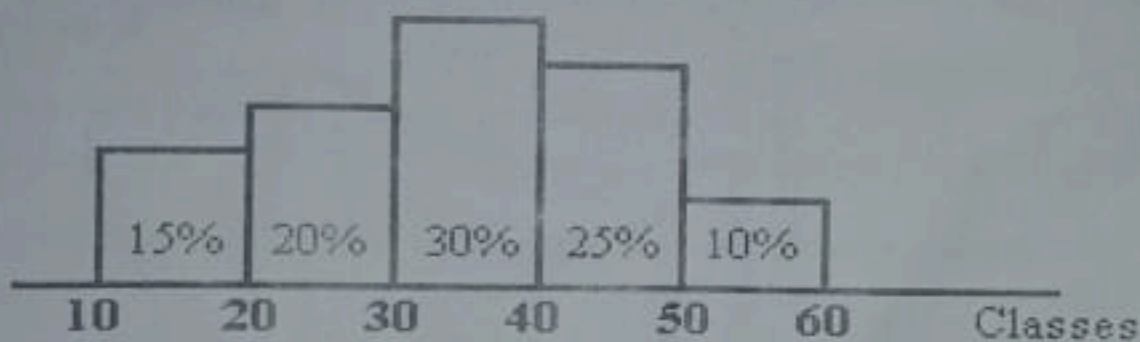
a) In a group of 800 guests at a given hotel, there are the following values regarding the length of stay at the hotel:

- Average = 9 days;
- 1st Quartile = 5 days;
- 3rd Quartile = 15 days;
- Coefficient of variation = 20%.

Using the above information, find:

- The number of guests who stayed more than 15 days [3 Marks]
- The number guests stayed between 5 and 15 days; [3 Marks]
- The standard deviation for the length of their stay. [4 Marks]

b) Given the following histogram with 80 observations, determine the mean, median, mode, standard deviation and 1st quartile of the distribution [10 Marks]



Question 3 [20 Marks]

How does Aaron Rodgers, quarterback for the 2011, Super Bowl winners, the Green Bay Packers, compare to Drew Brees, quarterback for the 2010 Super Bowl winners, the New Orleans Saints? The table below shows the number of completed passes for each athlete during the 2010 NFL football season:

Aaron Rodgers			Drew Brees		
19✓	21✓	7✓	27✓	37✓	25✓
19✓	15✓	25✓	28✓	34✓	29✓
34✓	27✓	19✓	30✓	27✓	35✓
12✓	22✓		33✓	29✓	22✓
27✓	26✓		24✓	23✓	
18✓	21✓		21✓	24✓	

- Calculate five-number summaries for the number of passes completed by both Aaron Rodgers and Drew Brees. [10 Marks]
- Construct box plots for the two sets of data. Are there any outliers? What do the box plots tell you about the shapes of the two distributions? [7 Marks]
- Write a short paragraph comparing the number of pass completions for the two quarterbacks. [3 Marks]

Question 4 [20 Marks]

- The 98.6-degree standard for human body temperature was derived by a German doctor in 1868. In an attempt to verify his claim, Mackowiak, Wasserman, and Levine took temperatures from 148 healthy people over 3 days. Allen Shoemaker derived a data set closely matching the one in Mackowiak's article and appears in the Journal of Statistics Education. The body temperatures for these 130 individuals are shown in the relative frequency histogram that follows.

$$\begin{bmatrix} 2 & 3 & 4 \\ 3 & 1 & 2 \\ -1 & 2 & 2 \end{bmatrix} R_1 \leftrightarrow R_1/2$$

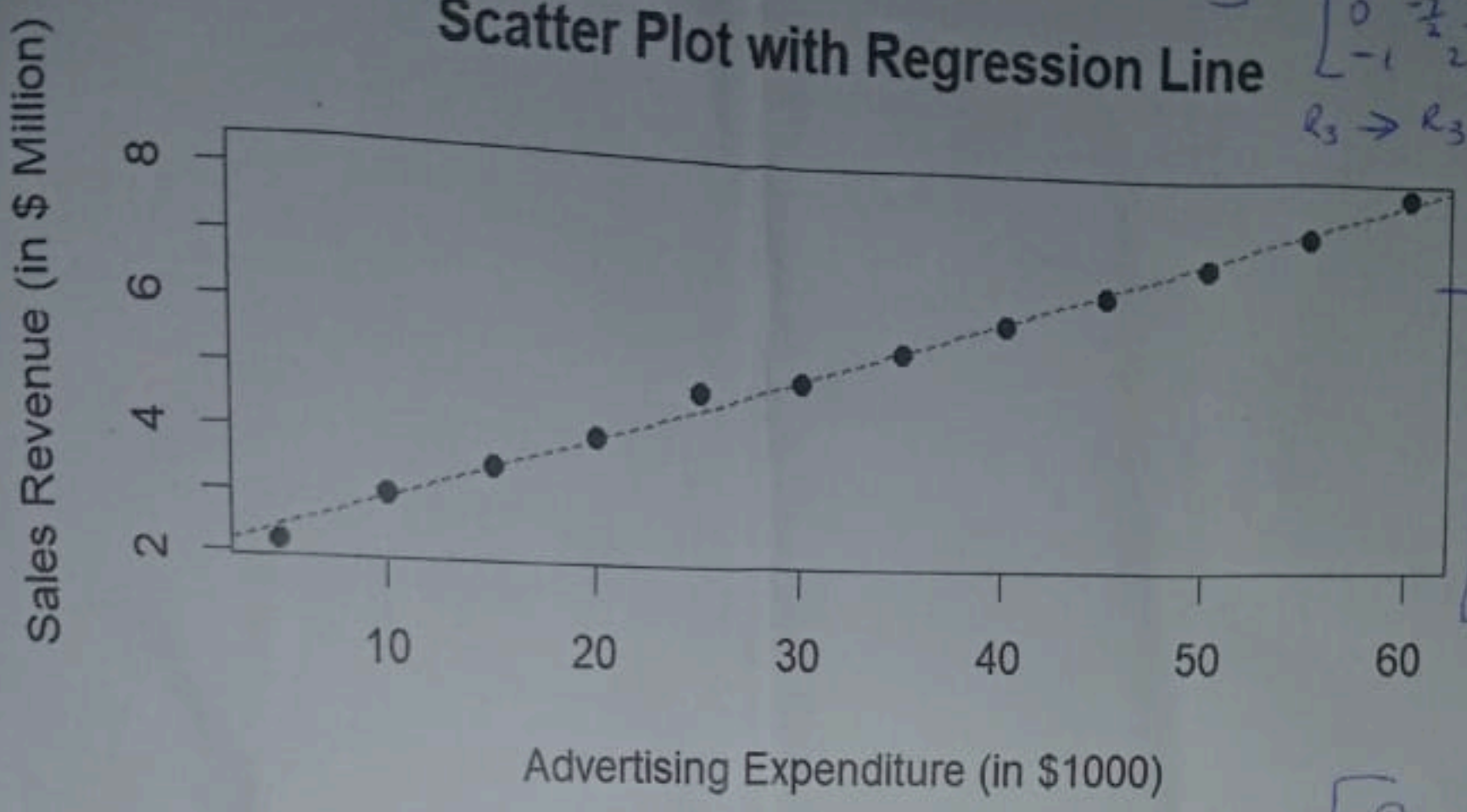
$$\begin{bmatrix} 1 & \frac{3}{2} & 2 \\ 0 & 1 & 2 \\ -1 & 2 & 2 \end{bmatrix} R_2 \rightarrow R_2 - 3$$

$$\begin{bmatrix} 1 & \frac{3}{2} & 2 \\ 0 & -\frac{1}{2} & -4 \\ -1 & 2 & 2 \end{bmatrix} R_3 \rightarrow R_3 +$$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & \frac{3}{2} \\ 0 & -\frac{7}{2} \\ 0 & 0 \end{bmatrix} R_3$$

Scatter Plot with Regression Line



Question 5 [20 Marks]

The table shows a verbal reasoning test score, X, and an English test score, Y, for each random sample of 8 learners who took both tests.

Learner	A	B	C	D	E	F	G	H
X	112	113	110	113	112	114	109	113
Y	69	65	75	70	70	75	68	76

- Differentiate between univariate and bivariate data [1 Mark]
- Calculate the value of Pearson's Correlation Coefficient between the scores in verbal reasoning test score, X, and an English test score, Y and interpret. [4 Marks]
- Using the computing formulas, write the regression equation. Interpret. [5 Marks]
- Sketch the scatterplot for the data and represent the best-fitting line in it. Compare these results with the ones in ii? [3 Marks]
- Supposing a learner had verbal reasoning test score of 130, what would the learner score in English test? [1 Mark]
- Determine the ANOVA table and explain its importance in the Regression Analysis [6 Marks]